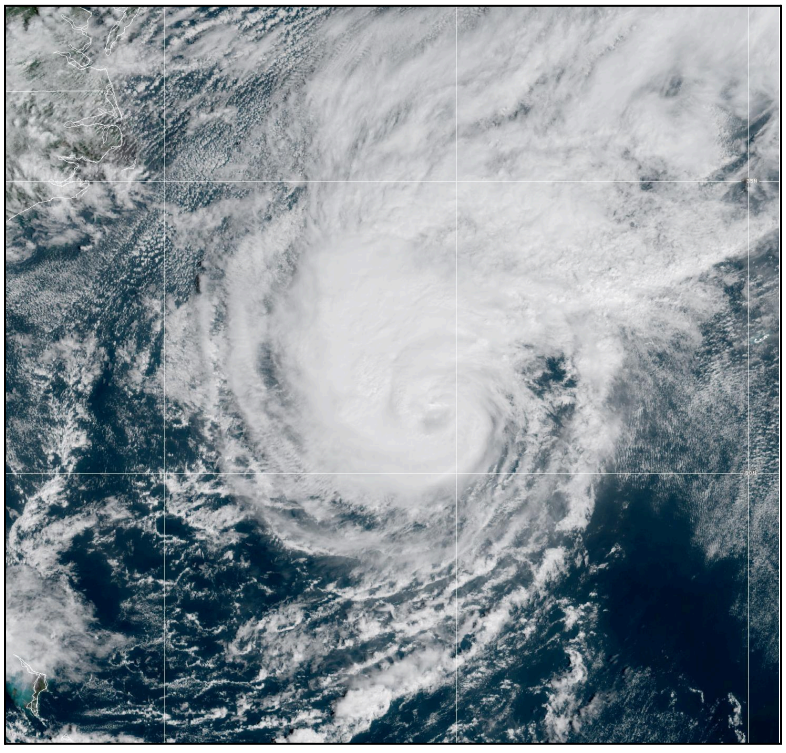
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT

HURRICANE IMELDA

(AL092025)

28 September–2 October 2025

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National Hurricane Center
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GOES-EAST GEOCOLOR SATELLITE IMAGE OF HURRICANE IMELDA AT 1500 UTC 1 OCTOBER 2025. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Imelda moved through the northwestern and central Bahamas as a tropical storm before turning east-northeastward and passing just south of Bermuda as a category 1 hurricane (on the Saffir-Simpson Hurricane Wind Scale). Heavy rainfall produced by Imelda and its predecessor tropical wave over the Greater Antilles resulted in 3 fatalities.

Hurricane Imelda

28 September–2 October 2025

Table of Contents

SYNOPTIC HISTORY....3
METEOROLOGICAL STATISTICS....4
Winds and Pressure....5
Storm Surge....6
Rainfall and Flooding....6
Greater Antilles....6
Florida and the Southeastern United States....6
Bahamas....7
Bermuda....7
CASUALTY AND DAMAGE STATISTICS....7
Southeastern United States....7
Greater Antilles....8
The Bahamas....8
Bermuda....8
FORECAST AND WARNING CRITIQUE....8
Genesis....8
Track....9
Intensity....10
Wind Watches and Warnings....10
IMPACT-BASED DECISION SUPPORT SERVICES & PUBLIC COMMUNICATION....10
ACKNOWLEDGEMENTS....11
TABLES....12
FIGURES....26

Hurricane Imelda

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SYNOPTIC HISTORY

Imelda originated from a tropical wave that moved off the west coast of Africa on 16 September. The wave produced some disorganized showers and thunderstorms as it moved westward at about 15 kt and passed over the Cabo Verde Islands on 17 September. Convective activity diminished during the next several days as the wave moved through a drier mid-level environment over the eastern and central tropical Atlantic. Showers and thunderstorms then gradually increased, but remained disorganized, on 21–22 September as the wave approached the Leeward Islands. Moving west-northwestward, the convectively active wave crossed the Leeward Islands on 23 September and then brought gusty winds and heavy rainfall to Puerto Rico and the Virgin Islands while crossing the region on 24 and early 25 September. Showers and thunderstorms began showing increased signs of organization later on 25 and 26 September while the disturbance passed over Hispaniola and eastern Cuba. However, satellite images and scatterometer data indicated the tropical wave still lacked a well-defined, closed circulation center. Given the high likelihood of development and the threat of tropical storm conditions to land areas, the NHC initiated advisories on Potential Tropical Cyclone Nine at 2100 UTC 26 September. At that time, the broad low-level vorticity maximum was located about 50 n mi north-northwest of the eastern tip of Cuba. The “best track” is given in Fig. 1, with the wind and pressure histories shown in Figs. 2 and 3, respectively. The best track positions and intensities are listed in Table 1[1].

The disturbance moved slowly northwestward on 26–27 September along the southwestern extent of a subtropical ridge over the western Atlantic. During this time, the convection did not become significantly better organized, and dropsonde data from Hurricane Hunter aircraft indicate the system remained an open surface trough between eastern Cuba and the southeastern Bahamas. The low-level circulation became better defined in satellite images late on 27 September, and convection was sufficiently organized to mark the formation of a tropical depression by 0000 UTC 28 September, when it was centered about 240 n mi south of Great Abaco Island in the northwestern Bahamas. Convective banding became more pronounced over the eastern side of the circulation later that day, and the depression strengthened into Tropical Storm Imelda by 1800 UTC 28 September when it was located about 155 n mi south of Great Abaco. Although the vortex was tilted with height due to moderate southerly shear (Fig. 4), Imelda gradually strengthened over 30°C waters near the Bahamas while it moved northward along the western side of the subtropical ridge. Imelda made landfall as a 45-kt tropical storm on Great Abaco around 1200 UTC 29 September.

The strengthening of Imelda continued later on 29 September while the system moved over 28–29°C waters north of the Bahamas (Fig. 5), but then the intensity leveled off early on 30

[1] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year’s storms are located in the *btk* directory, while previous years’ data are located in the *archive* directory.

September as continued shear and intrusions of dry air disrupted Imelda's convective organization and briefly left the low-level center exposed (Fig. 6, left). Meanwhile, the northward progress of Imelda slowed as nearby Hurricane Humberto[2], located a little more than 400 n mi to the east of Imelda, weakened the western Atlantic subtropical ridge. In response, Imelda turned sharply east-northeastward within the southwesterly flow ahead of an approaching mid-latitude trough over the eastern United States. Intensification resumed later on 30 September as new inner core convection consolidated around the center of Imelda, and it became a hurricane by 1200 UTC that day about 150 n mi north of Great Abaco. Enhanced upper-level divergence associated with the upper trough and warm sea-surface temperatures supported continued strengthening of the cyclone on 1 October as Imelda accelerated east-northeastward toward Bermuda. A ragged eye became apparent in satellite images (Fig. 6, right), and Imelda reached a peak intensity of 80 kt by 1200 UTC 1 October, when it was centered about 355 n mi west-southwest of Bermuda.

Extratropical transition began as Imelda continued to interact with the upper trough on its approach to Bermuda. Intrusions of dry air started to erode the inner core, and convection became displaced to the north of the cyclone center. In addition, the wind field grew in size and became asymmetric, with a narrow band of strong winds wrapping around the northern and western sides of the circulation as Imelda began to interact with a nearby front. The center of Imelda passed about 5–10 n mi south of Bermuda early on 2 October, with Imelda making its closest approach to the island around 0300 UTC as an 80-kt hurricane (Fig. 7). Synthetic aperture radar data from later that morning (Fig. 8) indicate Imelda became a frontal low and completed its extratropical transition by 1200 UTC 2 October, while centered about 200 n mi east-northeast of Bermuda. The powerful cyclone continued east-northeastward through early 3 October, then turned northward and briefly strengthened into a hurricane-force extratropical low by 0000 UTC 4 October. Then, the system accelerated northeastward over the northern Atlantic and gradually weakened over the next couple of days before being absorbed by another frontal system by 0600 UTC 6 October.

METEOROLOGICAL STATISTICS

Observations in Imelda (Figs. 2 and 3) include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Imelda.

[2] Hagen, Andrew B. "Tropical Cyclone Report: Hurricane Humberto." National Oceanic and Atmospheric Administration / National Weather Service / National Hurricane Center, 11 Dec. 2025, https://www.nhc.noaa.gov/data/tcr/AL082025_Humberto.pdf

Aircraft observations include flight-level and dropwindsonde data from 11 flights of the 53rd Weather Reconnaissance Squadron of the U.S. Air Force Reserve Command and 16 flights of the NOAA Hurricane Hunters (Fig. 9). Of these 16 NOAA flights, 8 were synoptic surveillance missions by the Gulfstream-IV jet. Several of these missions successfully performed circumnavigations of two systems (93L/Humberto and 94L/Imelda) on the same flight, which was the first time that this strategy has been employed. In total, the Air Force Reserve and NOAA Hurricane Hunters provided 42 center fixes for Imelda from 27 September–1 October.

Ship reports of tropical-storm-force winds associated with Imelda are given in Table 2, and selected surface observations from land stations and data buoys are given in Table 3.

Winds and Pressure

The estimated peak intensity of Imelda (80 kt) on 1 October is based on a blend of aircraft data and satellite intensity estimates. An Air Force Reserve Hurricane Hunter flight on 1 October measured peak 700-mb flight-level winds of 98 kt at 2140 UTC, which would typically correspond to a surface intensity of 88 kt using a standard 90% adjustment factor. However, Imelda was in the early stages of extratropical transition at this time, and layer-averaged dropsonde winds from that day suggest a standard reduction was likely not representative. A 1458 UTC eyewall dropsonde recorded mean boundary layer (0-500 m) and WL150 (0-150 m layer average) winds of 90 kt, which both reduce to an intensity of around 75 kt. Dropsonde data from the following mission supported even weaker surface-adjusted winds. Subjective Dvorak estimates from TAFB and SAB, as well as objective ADT and AiDT estimates from UW-CIMSS, peaked at 77 kt on 1 October. A blend of the various datasets supports a peak intensity of 80 kt.

The estimated minimum central pressure of 966 mb at 1800 UTC 1 October is based on aircraft data. A 1453 UTC center dropsonde from the Air Force Reserve Hurricane Hunters recorded a surface pressure of 967 mb with 16-kt surface winds, and a 1634 UTC center dropsonde measured a surface pressure of 968 mb with 20-kt surface winds.

Some of the northwestern and central Bahamas experienced tropical storm conditions on 28–29 September as Imelda moved over the region. A WeatherSTEM station on Blue Lagoon Island near Nassau measured a sustained wind of 36 kt and a gust of 41 kt at 1808 UTC 28 September. The next day, an unofficial weather station on Hog Cay in the northwestern Bahamas reported a sustained wind of 50 kt and a gust of 72 kt as the center of Imelda passed nearby.

A few coastal locations in east-central Florida experienced tropical-storm-force gusts on 29–30 September as the outer western portion of Imelda's circulation grazed the coastline. A gust of 37 kt was reported at Parrish Park in Titusville, and a 35-kt gust was measured in New Smyrna Beach.

Imelda brought hurricane conditions to Bermuda early on 2 October as the center of the storm passed just south of the island. An elevated weather station at the National Museum of Bermuda (45 m above sea level) measured a sustained wind of 67 kt at 0359 UTC and a gust of 86 kt. Elsewhere, a sustained wind of 41 kt and a gust of 57 kt were reported at L.F. Wade

International Airport (TXKF) on the northeastern part of the island. The minimum pressure recorded on the island was 974.7 mb at the National Museum of Bermuda.

Storm Surge[3]

For Bermuda, the highest measured storm surge reached 2.25 feet above normal tide levels along the north coast of St. George's Island on 2 October. This observed peak occurred at low tide, yielding only minor coastal inundation (i.e., 0.56 feet above Mean Higher High Water [MHHW]). However, elevated water levels persisted into the following high tide cycle, resulting in a peak storm tide (combination of storm surge and astronomical tide) of 1.09 ft above MHHW. It should be noted that peak water level observations are not available along the south coast of Bermuda, where the strongest onshore winds are likely to have occurred.

Rainfall and Flooding

Greater Antilles

The tropical wave that spawned Imelda produced heavy rainfall across much of the Greater Antilles as it moved across the region. The southern part of Puerto Rico generally received 2–4 inches of rainfall from 24–26 September, with a storm total maximum of 5.69 inches at a CoCoRaHS station near Cabo Rojo (PR-CR-13). These heavy rains caused flash flooding and some landslides, especially across southern portions of the island. Flooding was also reported along the Blanco, Cotui, and Viejo Rivers. Significant flash and river flooding occurred in the Dominican Republic (Fig. 10) as 4–8 inches (~100–200 mm) of rain with locally higher amounts fell across southern and eastern portions of the country from 23–27 September, with a storm total maximum of 21.13 inches (536.6 mm) in Oviedo. Media reports indicate heavy rainfall and flooding also occurred in Haiti.

An extended period of heavy rainfall occurred in eastern Cuba from 26–30 September, with numerous stations in Guantánamo and Santiago de Cuba receiving more than 12 inches (~ 300 mm) of rain from Imelda and its precursor disturbance. A storm total maximum rainfall of 26.87 inches (682.4 mm) was reported at Hatibonico, and 22.00 inches (559.0 mm) fell in Aserradero. These heavy rains triggered widespread flash flooding, landslides, and river flooding that required some water rescues.

Florida and the Southeastern United States

Outer rainbands from Imelda produced 1–3 inches of rain along portions of the east coast of Florida, with a storm total maximum of 4.35 inches measured at a CoCoRaHS station near Ormond Beach (FL-VL-104). The interaction of peripheral moisture from Imelda with a

[3] Several terms are used to describe water levels due to a storm. **Storm surge** is defined as the abnormal rise of water generated by a storm, over and above the predicted astronomical tide, and is expressed in terms of height above normal tide levels. Because storm surge represents the deviation from normal water levels, it is not referenced to a vertical datum. **Storm tide** is defined as the water level due to the combination of storm surge and the astronomical tide, and is expressed in terms of height above a vertical datum, i.e. the North American Vertical Datum of 1988 (NAVD88) or Mean Lower Low Water (MLLW). **Inundation** is the total water level that occurs on normally dry ground as a result of the storm tide, and is expressed in terms of height above ground level. At the coast, normally dry land is roughly defined as areas higher than the normal high tide line, or Mean Higher High Water (MHHW).

stalled front near the southeastern U.S. coast resulted in heavy rainfall across portions of North and South Carolina.

Bahamas

Imelda produced flooding rains across the northwestern and central Bahamas from 27–30 September. The heaviest rainfall occurred on the island of New Providence, with a storm total maximum of 14.87 inches of rain measured at the Lynden Pindling International Airport (MYNN) near Nassau. Other locations on that island received 6–10 inches of rainfall during the 3-day period. In Great Abaco, 7.76 inches of rain was reported at a CoCoRaHS station (BHS-CO-4) near Cherokee. Elsewhere, storm total rainfall amounts of 3–6 inches were reported on other islands including San Salvador, Great Harbour Cay, and Eleuthera. Media reports indicate flooding occurred on the islands of New Providence, Abaco, and San Salvador.

Bermuda

Imelda was undergoing extratropical transition as it passed near Bermuda, and the storm's fast forward speed limited the duration of rainfall that occurred on the island. The L.F. Wade International Airport (TXKF) reported 1.30 inches of rain. No significant flooding was reported.

CASUALTY AND DAMAGE STATISTICS

Imelda is responsible for 2 direct fatalities[4] in the Santiago de Cuba province of eastern Cuba due to freshwater flooding, including a 60-year-old man who died in a landslide following heavy rainfall. In Puerto Rico, a 36-year-old man drowned when his car was swept away by floodwaters after heavy rainfall caused the Yauco River to overflow its banks on 24 September[5]. This fatality is attributed to the parent tropical wave from which Imelda formed. There are no official damage estimates available at the time of this report.

Southeastern United States

Large swells from both Imelda and distant Hurricane Humberto caused an extended period of rough surf and damaging waves along the southeastern U.S. coast. In Florida, some minor to moderate beach erosion was noted by officials in Brevard and Volusia Counties. Farther north in the Outer Banks of North Carolina, more significant beach erosion caused by battering waves led to 6 unoccupied homes in Buxton collapsing into the ocean.[6]

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[4] Deaths occurring as a direct result of the forces of the tropical cyclone are referred to as "direct" deaths. These would include those persons who drowned in storm surge, rough seas, rip currents, and freshwater floods. Direct deaths also include casualties resulting from lightning and wind-related events (e.g., collapsing structures). Deaths occurring from such factors as heart attacks, house fires, electrocutions from downed power lines, vehicle accidents on wet roads, etc., are considered "indirect" deaths.

[5] https://wapa.tv/noticias/locales/profundo-dolor-joven-muerto-en-inundaci-n-de-yauco-era-due-o-de-panader-a/article_838ec8f8-e16e-48ee-bd29-bc490916231d.html

[6] https://apnews.com/article/outer-banks-homes-collapse-hurricane-beachside-north-carolina-1c277f5ea323e2464e0d83a130cdcea7

Greater Antilles

Imelda and its predecessor tropical wave produced damaging floods across portions of Puerto Rico, the Dominican Republic, Haiti, and Cuba[7]. In Puerto Rico, several landslides blocked roads, and other roads were inundated by swollen rivers. Some downed trees and utility poles were also reported. In the Dominican Republic, 2 homes were destroyed and almost 1,000 homes were damaged by flooding. River flooding temporarily cut off some communities, and gusty winds over the saturated ground led to some downed trees that blocked roads and damaged a couple of homes. Elsewhere, flooding affected 35 communities in Haiti and caused significant crop damage. In eastern Cuba, flooding and landslides cut off 17 communities in the Santiago de Cuba province affecting more than 24,000 people. Over 18,000 people were evacuated in the Guantánamo province, and damaged roads and bridges were reported where numerous landslides and extensive flash flooding occurred.

The Bahamas

No major structural damage was reported in the Bahamas. Significant flooding occurred in parts of Nassau and surrounding communities on the island of New Providence, including Carmichael, Pinewood Gardens, Gamble Heights, Fox Hill, Bel Air Estates, and West Grove[8]. Numerous homes and buildings were inundated by floodwaters, including about 100 homes in Carmichael. Some residents and vehicles were stranded by floodwaters, and several water rescues were performed. More than a dozen schools in New Providence, Grand Bahama, and Abaco were closed due to flooding concerns. Gusty winds downed some power lines, with widespread power outages reported in Abaco and more than 1,500 customers affected in New Providence.[9]

Bermuda

No significant damage or casualties occurred in Bermuda. Strong winds downed some trees and power lines across the island, and about 18,000 customers lost power at the height of the event. In Pembroke Parish, the roof of a former hotel was ripped off, and an exterior wall collapsed, which damaged a nearby vehicle (Fig. 11).[10]

FORECAST AND WARNING CRITIQUE

Genesis

The genesis of Imelda was anticipated well in advance, although the system formed much later than originally forecast (Table 4). The wave from which Imelda developed was originally introduced in the 7-day Tropical Weather Outlook (TWO) shortly after it emerged off the west coast of Africa, 282 h (11.75 days) prior to genesis. A low 2-day chance of formation

[7] https://apnews.com/article/hurricane-imelda-humberto-bermuda-c6760b885b541c2154dcb1b5b9bcf163
[8] https://www.thenassauguardian.com/news/flooded-out/article_5c819cc7-d3fb-44d3-a6ea-a39183462553.html
[9] https://www.tribune242.com/news/2025/sep/30/imelda-leaves-behind-flooding-in-its-wake/
[10] https://www.royalgazette.com/weather/news/article/20251003/storm-takes-out-wall-at-former-hotel/

was introduced 264 h (11 days) prior to formation. However, the wave was embedded in a dry mid-level environment and failed to develop during the following couple of days, and the system was dropped from the TWO on 18 September. Several days later, the tropical wave was reintroduced into the 7-day and 2-day TWO with a low chance of formation 150 h (6.25 days) prior to genesis. The 7-day probabilities were raised to the medium and high categories 132 h and 96 h before formation, respectively. For the 2-day TWO, the system reached the medium category 72 h and the high category 60 h before genesis occurred. The genesis forecasts for Imelda were challenging due to inconsistent global model and ensemble signals as to when development might occur, likely related to the hostile environmental conditions that the wave encountered within the Main Development Region during the climatological peak of the season.

A verification of NHC Graphical TWO (GTWO) genesis forecasts for Imelda is shown in Figure 12. The struggles to pinpoint the timing of formation also made it difficult to predict exactly where Imelda would form, as shown by the 43% overall GTWO hit rate. Early NHC genesis forecasts indicated a low chance of formation over the eastern and central tropical Atlantic, but this did not occur due to the marginal environmental conditions. After the wave was reintroduced into the GTWO, the NHC genesis forecasts generally highlighted the southwestern Atlantic Ocean for formation, but most of the medium probability areas were east of where Imelda ultimately formed. The high category areas all correctly captured the location of genesis.

Track

A verification of NHC official track forecasts (OFCL) for Imelda is given in Table 5a. Official track forecast errors were comparable to or lower than the mean official errors for the previous 5-yr period from 12–60 h, and greater than the mean errors at 72 h. For the two verifying 96-h forecasts, OFCL errors were over 3.5 times larger than the 5-yr means. A homogeneous comparison of the official track errors with selected guidance models is given in Table 5b. The OFCL forecasts outperformed many of the individual models and consensus aids from 12–72 h, although OFCL was generally bested by the UKMET (UKMI) and Google DeepMind ensemble (GDMI) during this period. At 96 h, OFCL had larger errors than most of the guidance, albeit for only 2 verifying forecasts.

The close proximity of Humberto and Imelda and the potential for binary interaction made early track forecasts very challenging. In particular, there was a bifurcation in the long-range track guidance available when advisories were initiated (Fig. 13). While some models (GFS, HAFS-A/B, ECMWF) showed a close approach to or even a landfall along the southeastern U.S. coast, others (COAMPS-TC and UKMET) favored a sharp eastward turn farther offshore. These diverging track solutions were also noted in the various global ensembles (not shown). Humberto and pre-Imelda were at roughly the same latitude on 26–27 September, but Imelda developed more gradually and moved northward more slowly than Humberto on 28–30 September (Fig. 14). As a result, Humberto eroded the steering ridge over the western Atlantic, which kept Imelda farther offshore than originally forecast. While the official NHC forecast never explicitly predicted landfall, early NHC forecasts showed a much closer approach to land than what ultimately occurred before trending eastward in response to large shifts in the track guidance (Fig. 15). Once the advancing upper-level trough over the eastern U.S. became the predominant steering feature (due to the slower storm motion of Imelda), the tropical cyclone turned sharply east-northeastward away from the U.S. and accelerated toward

Bermuda. Notably, the UKMET was among the first dynamical models to show Imelda making this sharp eastward turn away from the U.S. coastline (Fig. 13). The NHC track forecasts consistently showed Imelda making a close approach to Bermuda for several days prior to the hurricane impacting the island (Fig. 16).

Intensity

A verification of NHC official intensity forecasts (OFCL) for Imelda is given in Table 6a. The NHC intensity forecasts for Imelda were very good, with OFCL errors comparable to the 5-yr mean at 12 h and lower than the 5-yr means at 24–96 h. Figure 17 shows all of the NHC intensity forecasts for Imelda compared to the best track. The official forecasts generally captured the periods of strengthening and weakening reasonably well, although Imelda did not intensify as quickly as initially predicted. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 6b. The NHC intensity forecasts beat almost all of the individual models between 24–96 h, and OFCL bested HCCA, FSSE, and GDMI at a majority of verified forecast times.

Wind Watches and Warnings

Coastal wind watches and warnings associated with Imelda are given in Table 7. The NHC initiated potential tropical cyclone advisories at 2100 UTC 26 September on the precursor disturbance, which enabled the Bahamas to issue Tropical Storm Watches and Warnings before genesis occurred.

For the United States, a Tropical Storm Watch was issued for the east coast of Florida from the Palm Beach/Martin County Line to the Flagler/Volusia County Line at 1500 UTC 27 September. The watch was never upgraded to a warning since subsequent track forecasts shifted farther offshore. While no sustained winds of tropical storm force occurred along the coast, a few locations within the watch area did report tropical-storm-force gusts on 29–30 September.

IMPACT-BASED DECISION SUPPORT SERVICES & PUBLIC COMMUNICATION

The NHC began formal communication with emergency managers on 25 September before the disturbance that became Imelda reached the Bahamas. Thirteen decision support briefings were provided to emergency managers and coordinated through the Federal Emergency Management Agency (FEMA) Hurricane Liaison Team embedded at the NHC. These briefings included video-teleconferences with FEMA HQ, FEMA Region 4, the states of South Carolina and North Carolina, and the state and counties of Florida. Briefing support continued through 29 September, after which it became clear that Imelda would not threaten the United States. In addition, the NHC/TAFB provided seven briefings to the U.S. Coast Guard Southeast District from 27–30 September in support of their life-saving mission.

ACKNOWLEDGEMENTS

Meteorological data and storm summaries from the National Weather Service (NWS) Forecast Offices in Melbourne, Florida, and San Juan, Puerto Rico, NOAA National Data Buoy Center, Dominican Republic Institute of Meteorology (INDOMET), Cuban Meteorological Service (INSMET), Bahamas Department of Meteorology, and Bermuda Weather Service were used in this report. Jay Álamo (WPC International Desk Chief) helped to gather rainfall data. Dr. Lisa Bucci (NHC) created the aircraft reconnaissance summary graphic, Dr. Philippe Papin (NHC) created the Graphical TWO verification graphic, and Dr. Wallace Hogsett (NHC) created the track and intensity model verification figures. Michael Spagnolo (FEMA) contributed to the IDSS and Public Communication section, and Dr. Cody Fritz and Laura Alaka contributed to the Storm Surge section.

Table 1. Best track for Hurricane Imelda, 28 September–2 October 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
26 / 1800	20.9	74.9	1008	30	disturbance
27 / 0000	21.3	75.5	1006	30	"
27 / 0600	21.6	75.8	1006	30	"
27 / 1200	21.9	76.1	1005	30	"
27 / 1800	22.1	76.4	1005	30	"
28 / 0000	22.4	76.7	1004	30	tropical depression
28 / 0600	22.7	77.1	1003	30	"
28 / 1200	23.1	77.4	1001	30	"
28 / 1800	23.8	77.3	998	35	tropical storm
29 / 0000	24.6	77.1	997	35	"
29 / 0600	25.3	77.1	995	40	"
29 / 1200	26.4	77.1	989	45	"
29 / 1800	27.3	77.3	987	50	"
30 / 0000	28.0	77.3	985	55	"
30 / 0600	28.6	77.2	983	55	"
30 / 1200	28.9	76.9	981	65	hurricane
30 / 1800	29.2	76.0	979	70	"
01 / 0000	29.4	74.8	977	70	"
01 / 0600	29.9	73.1	973	75	"
01 / 1200	30.7	71.4	968	80	"
01 / 1800	31.2	69.2	966	80	"
02 / 0000	31.8	66.4	971	80	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
02 / 0600	32.6	63.2	974	80	"
02 / 1200	33.0	60.8	977	70	extratropical
02 / 1800	33.5	57.5	986	60	"
03 / 0000	34.5	53.4	990	55	"
03 / 0600	35.7	51.1	990	55	"
03 / 1200	37.3	49.3	986	55	"
03 / 1800	38.9	49.1	979	60	"
04 / 0000	39.7	49.0	978	65	"
04 / 0600	40.8	48.5	979	60	"
04 / 1200	42.0	47.5	981	55	"
04 / 1800	43.5	45.8	983	50	"
05 / 0000	45.3	44.0	984	45	"
05 / 0600	46.9	41.9	984	45	"
05 / 1200	48.6	39.6	983	50	"
05 / 1800	49.5	36.5	984	50	"
06 / 0000	50.3	33.0	988	45	"
06 / 0600					dissipated
29 / 1200	26.4	77.1	989	45	Landfall on Great Abaco, Bahamas
01 / 1800	31.2	69.2	966	80	minimum pressure & maximum winds

Table 2. Selected ship reports with winds of at least 34 kt for Hurricane Imelda, 28 September–2 October 2025. Due to varying anemometer heights, not all reports are considered representative of the storm's intensity.

Date/Time (UTC)	Ship call sign	Latitude (°N)	Longitude (°W)	Wind dir/ speed (kt)	Pressure (mb)
28 / 1500	9HA615	21.7	76.6	220 / 35	1006.3
28 / 1700	9HA615	21.4	75.9	210 / 45	1006.3
28 / 2000	9HA615	21.1	75.1	210 / 35	1007.3
29 / 0100	9HA615	20.4	73.5	200 / 35	1007.3
29 / 0200	9HA615	20.4	73.5	200 / 35	1007.3
29 / 1600	V7IE2	26.3	78.9	360 / 37	1008.0
30 / 1500	KOGH	32.6	75.4	040 / 38	1005.4
30 / 1500	C6EO4	33.0	72.1	030 / 35	1012.0
30 / 1600	V7IE2	26.3	78.8	300 / 37	1009.0
30 / 1700	C6EO4	33.3	71.9	020 / 40	1013.0
30 / 1800	WLIU	25.7	74.9	230 / 45	1004.8
30 / 2100	WLIU	26.1	75.5	220 / 45	1003.5
01 / 0000	WMCU	34.6	75.5	040 / 35	1008.4
01 / 0100	KOGH	32.4	74.2	050 / 42	1001.5
01 / 0400	KOGH	32.3	73.9	030 / 36	1000.3
01 / 0500	KOGH	32.3	73.8	020 / 45	999.2
01 / 0600	KOGH	32.2	73.7	030 / 40	999.3
01 / 0700	KOGH	32.2	73.6	020 / 45	997.8

Date/Time (UTC)	Ship call sign	Latitude (°N)	Longitude (°W)	Wind dir/ speed (kt)	Pressure (mb)
01 / 1000	KOGH	32.2	73.4	020 / 50	998.0
01 / 1200	C6EO4	35.0	71.0	360 / 36	1010.0
01 / 1500	KOGH	32.2	73.1	010 / 45	1003.3
01 / 1800	KOGH	32.3	72.9	360 / 38	1005.3
01 / 1800	WPAJ	36.4	74.1	040 / 35	1015.0
01 / 1900	WPAJ	36.2	74.3	030 / 35	1015.7
01 / 2100	WPAJ	35.8	74.5	020 / 36	1016.5
01 / 2300	KOGH	32.3	72.6	360 / 40	1010.0

Table 3. Selected surface observations for Hurricane Imelda, 28 September–2 October 2025. Some rainfall totals[c] include precipitation that fell before genesis during the tropical wave and potential tropical cyclone phases.

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Total rain (in)[c]
	Date/ time (UTC)	Pressure (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)	
Puerto Rico						
Cabo Rojo 3.9 S (PR-CR-13) (18.03N 67.15W)						5.69
Cabo Rojo 1.1 ENE (PR-CR-4) (18.10N 67.13W)						4.70
Ponce 3.5 W (PR-PC-8) (17.99N 66.66W)						4.52
Dominican Republic						
Oviedo						21.13
Enriquillo						17.48
Valle Nuevo						8.40
Arroyo Loro						8.31
La Romana Intl. AP						7.63
Cuba						
Guantánamo						
Guantánamo Bay (MUGM) (19.91N 75.16W)	27/2051	1008.2	28/1710	32	42	
Hatibonico						26.87
Embalse Los Asientos						20.61
Paraguay						17.28
San Antonio del Sur						16.85
Imías						15.08
Santiago de Cuba						
Aserradero						22.00
Canasí						20.83
Embalse Chalons						20.30
La Campana						20.20
Embalse Parada						17.03
The Bahamas						
Hog Cay (26.95N 77.60W)			29/1920	50	72	
Blue Lagoon Island WeatherSTEM (25.10N 77.27W)	29/0415	997.4	28/1808	36	41	
Lynden Pindling Intl. AP (MYNN) (25.04N 77.47W)			29/0332	23	38	14.87

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Total rain (in)[c]
	Date/ time (UTC)	Pressure (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)	
Grand Bahama Intl. AP (MYGF) (26.56N 78.70W)			30/0100	21	33	
New Providence (BHS-NP-13) Nassau 4.8 S (25.03N 77.34W)						9.94
Great Abaco (BHS-CO-4) Cherokee 5.7 NW (26.31N 77.10W)						7.76
San Salvador (BHS-SS-7) Cockburn Town 9.7 NE (24.10N 74.45W)						5.81
Great Harbour Cay (BHS-BY-1) Bullocks Harbor 10.6 NW (25.82N 77.94W)						3.79
Eleuthera (BHS-CE-8) Governor's Harbour 8.9 NW (25.26N 76.30W)						3.62
Florida						
Titusville Parrish Park North (XPAR) (28.63N 80.81W)			30/0310	28 (5 m, 1 min)	37	
New Smyrna Beach (XNSB) (29.05N 80.90W)			30/0452	27 (10 m, 1 min)	35	
Banana River – 520 (XCCB) (28.36N 80.65W)			30/0251	23 (5 m, 1 min)	36	
Ormond Beach 3.9 SW (FL-VL-104) (29.26N 81.15W)						4.35
New Smyrna Beach 3.9 W (FL-VL-77) (29.03N 80.98W)						3.69
Bermuda						
National Museum of Bermuda *elev. 75 ft* (32.33N 64.83W)	02/0256	974.7	02/0359	67 (21 m, 2 min)	86	
Airport AviMet – 30 (32.36N 64.67W)	02/0316	976.3	02/0434	43 (10 m, 2 min)	57	
L.F. Wade Intl. AP (TXKF) (32.36N 64.67W)	02/0315	977.3	02/0426	41 (10 min)	57	1.30
Esso Pier (32.37N 64.70W)	02/0306	979.0	02/0542	40 (8 m)	52	
Airport AviMet – 12 (32.37N 64.69W)	02/0315	976.8	02/0349	40 (10 m, 2 min)	51	
Ferry Reach Channel (FRCB6) (32.37N 64.69W)	02/0312	977.2				

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Total rain (in)[c]
	Date/ time (UTC)	Pressure (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)	
Offshore						
41010 – Canaveral East (28.88N 78.47W)	01/2157	1002.1	02/1310	37 (3.8 m, 1 min)	42	
41002 – South Hatteras (31.74N 74.96W)	01/0512	999.1	01/0457	35 (4 m, 1 min)	43	
41009 – Canaveral (28.51N 80.19W)	30/0720	1005.3	30/0304	31 (4 m, 1 min)	35	
41047 – NE Bahamas (27.56N 71.48W)	01/0700	1001.4				

[a] Date/time is for sustained wind when both sustained and gust are listed.
[b] If available, anemometer height above ground and sustained wind averaging periods are included in parentheses below the sustained wind speed.
[c] Storm total rainfall includes the following dates:
Puerto Rico: 24–26 September 2025
Dominican Republic: 23–27 September 2025
Cuba: 26–30 September 2025
Bahamas: 27–30 September 2025

Table 4. Number of hours in advance of Imelda's formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. The number in parentheses ( ) indicates the number of hours in advance of formation when the disturbance was re-introduced into the low category in the TWO. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	264 (150)	282 (150)
Medium (40%-60%)	72	132
High (>60%)	60	96

Table 5a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Imelda, 28 September–2 October 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	23.0	36.5	**42.4**	**35.8**	**54.6**	101.0	456.7	
OCD5	47.4	115.5	174.1	196.7	217.4	302.1	618.5	
Forecasts	16	14	12	10	8	6	2	
OFCL (2020-24)	23.0	34.3	45.8	58.7	73.5	89.8	128.7	185.4
OCD5 (2020-24)	45.1	95.7	150.9	203.1	252.7	295.4	366.2	426.6

Table 5b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Imelda, 28 September–2 October 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 5a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	21.8	34.3	42.4	35.2	54.6	101.0	456.7	
OCD5	48.0	110.7	169.8	177.0	217.4	302.1	618.5	
GFSI	32.2	46.6	**41.0**	46.7	62.5	114.0	**328.6**	
HWFI	30.1	45.2	60.0	89.6	141.6	216.2	**448.7**	
HMNI	29.5	56.5	79.8	90.9	121.2	209.2	603.9	
HFAI	27.6	43.1	51.8	72.8	95.3	134.4	494.9	
HFBI	29.0	51.6	59.5	70.4	94.9	149.2	503.8	
EMXI	**21.7**	**30.7**	**35.6**	45.5	70.6	151.7	**445.4**	
UKMI	**21.1**	**24.2**	**27.1**	**33.0**	**50.7**	110.6	**221.8**	
NVGI	38.2	75.3	130.4	174.9	227.3	234.0	**375.3**	
CMCI	**21.7**	**32.1**	51.5	71.4	97.5	105.3	**129.9**	
CTCI	26.8	46.3	68.5	62.7	73.8	137.1	**429.7**	
TVCA	25.5	37.6	43.5	45.5	68.2	137.5	**414.1**	
TVCX	24.9	36.7	**41.5**	46.1	68.3	139.4	**417.7**	
GFEX	26.5	36.7	**37.2**	39.0	56.9	130.5	**385.2**	
TVDG	25.0	35.9	**39.0**	41.0	60.0	128.7	**380.6**	
HCCA	21.9	35.6	43.5	43.7	**53.8**	108.1	**298.6**	
FSSE	24.7	36.5	**37.9**	39.2	**50.9**	102.4	**328.4**	

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
AEMI	27.1	37.4	44.8	46.0	**44.8**	**67.4**	**338.4**	
GDMI	**18.7**	**31.8**	**38.0**	44.3	**40.4**	**56.8**	**261.2**	
TABS	59.5	116.9	164.2	181.8	192.4	165.0	**108.5**	
TABM	35.9	68.5	102.7	161.0	218.1	285.7	**294.0**	
TABD	26.6	41.0	68.9	119.7	176.6	229.6	**261.6**	
Forecasts	14	12	11	9	8	6	2	

Table 6a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Imelda, 28 September–2 October 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.3	**5.0**	**5.0**	**4.0**	**3.8**	**1.7**	**7.5**	
OCD5	4.6	5.4	7.5	11.4	16.2	22.5	27.0	
Forecasts	16	14	12	10	8	6	2	
OFCL (2020-24)	5.1	7.3	8.6	10.0	10.5	10.9	12.4	13.6
OCD5 (2020-24)	6.8	10.6	13.8	16.5	17.9	19.2	21.4	19.9

Table 6b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Imelda, 28 September–2 October 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 6a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.4	5.0	5.0	4.4	3.8	1.7	7.5	
OCD5	**4.4**	5.1	6.8	11.2	16.2	22.5	27.0	
HWFI	5.4	5.2	5.0	5.9	4.8	6.7	12.5	
HMNI	**3.9**	6.7	7.4	7.7	10.5	8.5	17.5	
HFAI	**4.1**	6.1	6.2	4.6	6.8	7.3	16.0	
HFBI	**4.9**	8.2	7.1	7.4	8.5	6.5	**4.0**	
DSHP	**4.6**	6.9	9.4	9.7	7.2	4.7	14.5	
LGEM	**5.2**	5.2	7.1	9.3	9.6	12.5	20.0	
ICON	**3.9**	5.0	5.4	7.6	5.1	4.3	9.0	
IVCN	**3.9**	**4.6**	5.5	6.9	5.6	4.7	8.0	
IVDR	**4.0**	**4.8**	5.5	6.7	5.9	4.5	**6.5**	
CTCI	**4.7**	5.8	6.4	8.0	7.4	6.7	**2.0**	
GFSI	**4.0**	6.7	8.5	10.7	11.0	9.5	10.0	
EMXI	5.8	9.1	13.0	17.6	20.2	20.2	17.5	
HCCA	5.6	5.5	6.0	**4.1**	4.1	4.3	**3.5**	
FSSE	**5.1**	6.4	5.7	5.0	**3.0**	4.5	**2.0**	
GDMI	**4.9**	6.7	6.2	7.9	7.4	6.3	**2.5**	
Forecasts	14	12	11	9	8	6	2	

Table 7. Watch and warning summary for Hurricane Imelda, 28 September–2 October 2025.

Date/Time (UTC)	Action	Location
26 / 2100	Tropical Storm Warning issued	Central Bahamas, including Cat Island, the Exumas, Long Island, Rum Cay, and San Salvador
26 / 2100	Tropical Storm Watch issued	Portions of the Northwestern Bahamas, including Eleuthera, New Providence, the Abacos, Berry Islands, and Grand Bahama Island
27 / 0900	Tropical Storm Watch upgraded to Tropical Storm Warning	Portions of the Northwestern Bahamas, including Eleuthera, New Providence, the Abacos, Berry Islands, Andros Island, and Grand Bahama Island
27 / 1500	Tropical Storm Watch issued	East coast of Florida from the Palm Beach/Martin County Line to the Flagler/Volusia County Line
28 / 2100	Tropical Storm Watch discontinued	East coast of Florida from the Palm Beach/Martin County Line to the Flagler/Volusia County Line
29 / 0900	Tropical Storm Warning discontinued	Central Bahamas and portions of the Northwestern Bahamas, including New Providence, Berry Islands, and Andros Island
29 / 2100	Hurricane Watch issued	Bermuda
29 / 2100	Tropical Storm Warning discontinued	Northwestern Bahamas, except Grand Bahama and Great Abaco Island
30 / 0300	Tropical Storm Warning discontinued	Northwestern Bahamas (all)
30 / 1500	Hurricane Watch upgraded to Hurricane Warning	Bermuda
02 / 0900	Hurricane Warning changed to Tropical Storm Warning	Bermuda
02 / 1500	Tropical Storm Warning discontinued	Bermuda

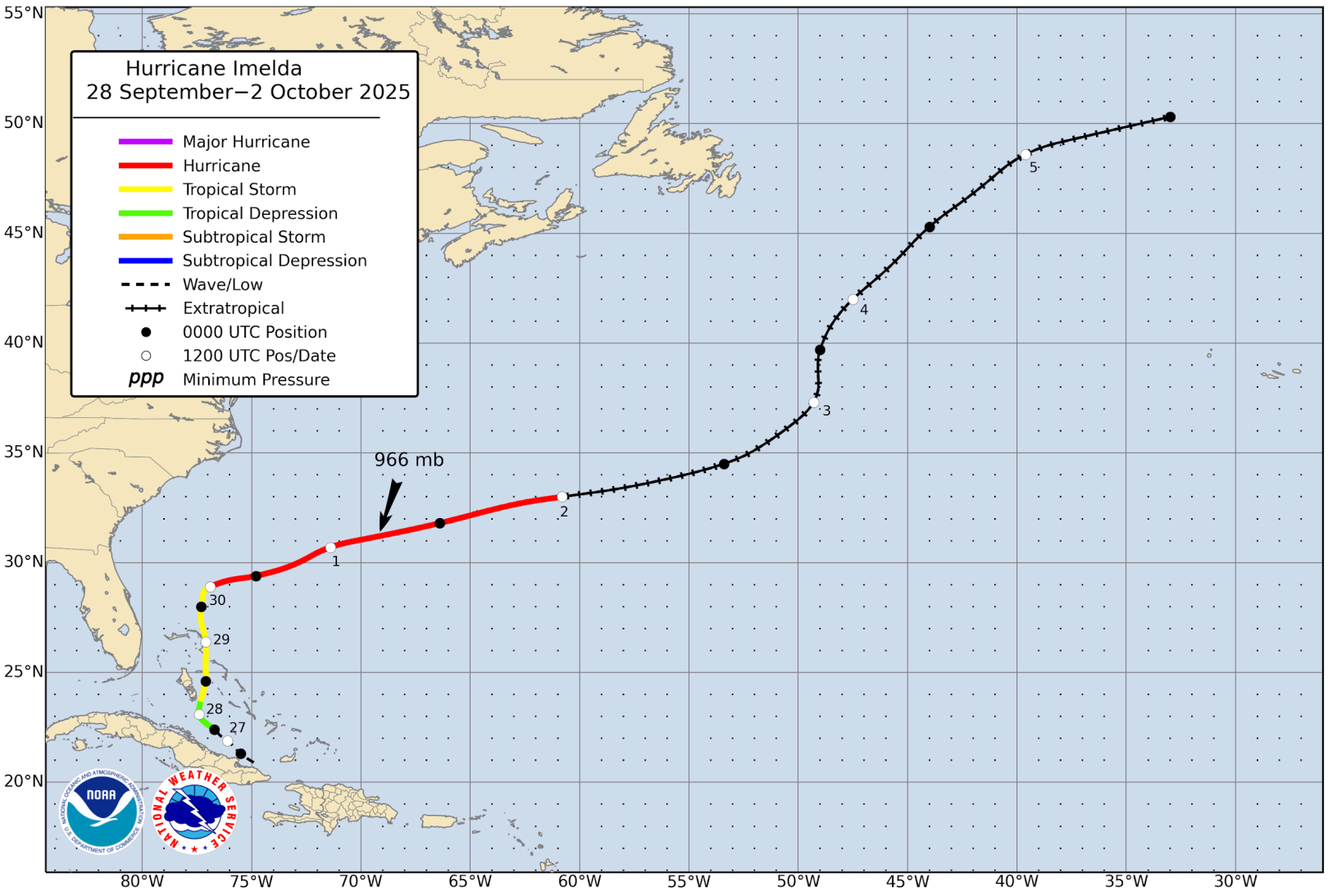


Figure 1. Best track positions for Hurricane Imelda, 28 September–2 October 2025. Tracks during the extratropical stage are partially based on analyses from the NOAA Ocean Prediction Center.

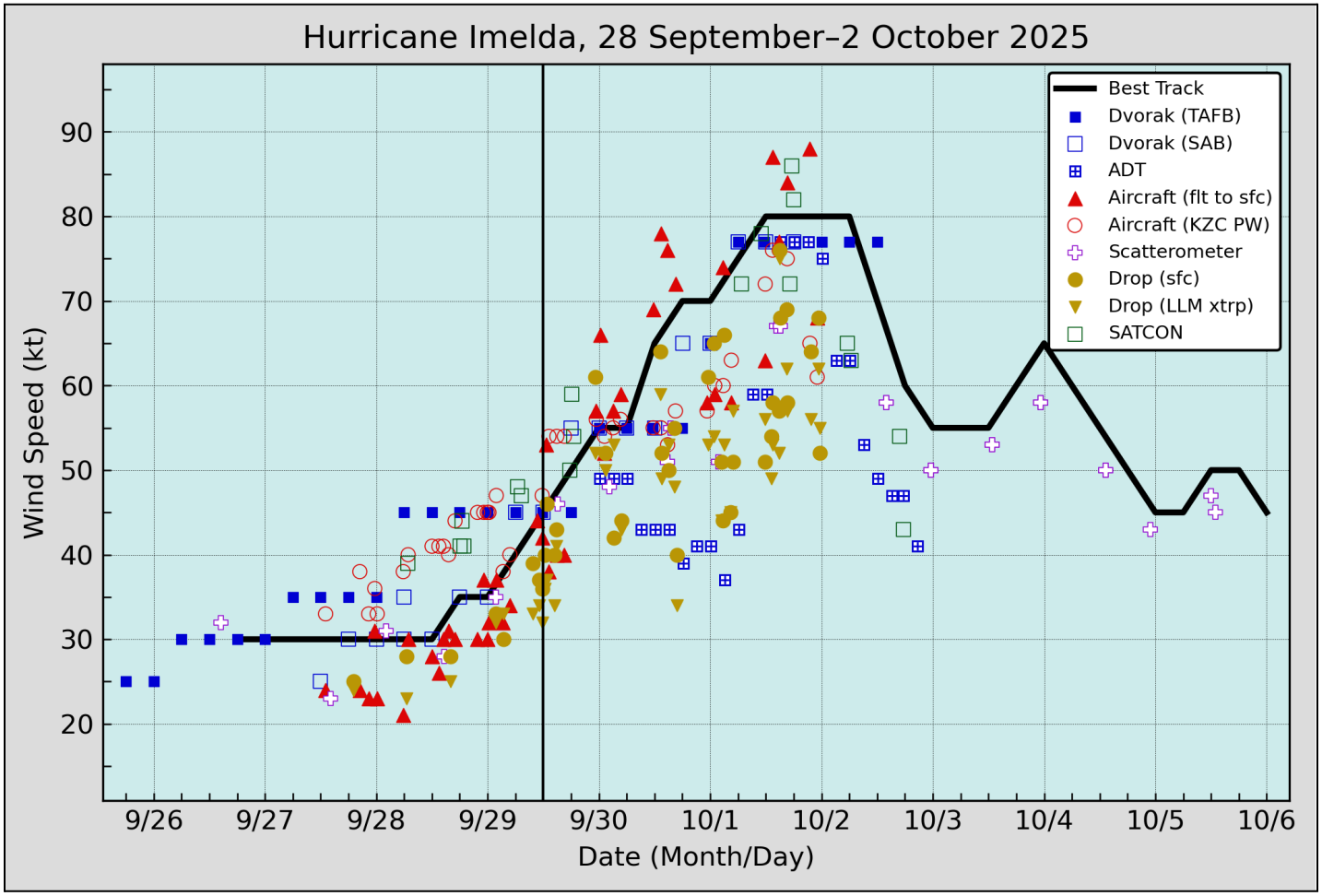


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Imelda, 28 September–2 October 2025. Aircraft observations have been adjusted for elevation using 90%, 80%, and 75% adjustment factors for observations from 700 mb, 850 mb, and 925 mb, respectively. Dropwindsonde observations include actual 10 m winds (sfc), as well as surface estimates derived from the mean wind over the lowest 150 m of the wind sounding (LLM). Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC, and the solid vertical line corresponds to landfall.

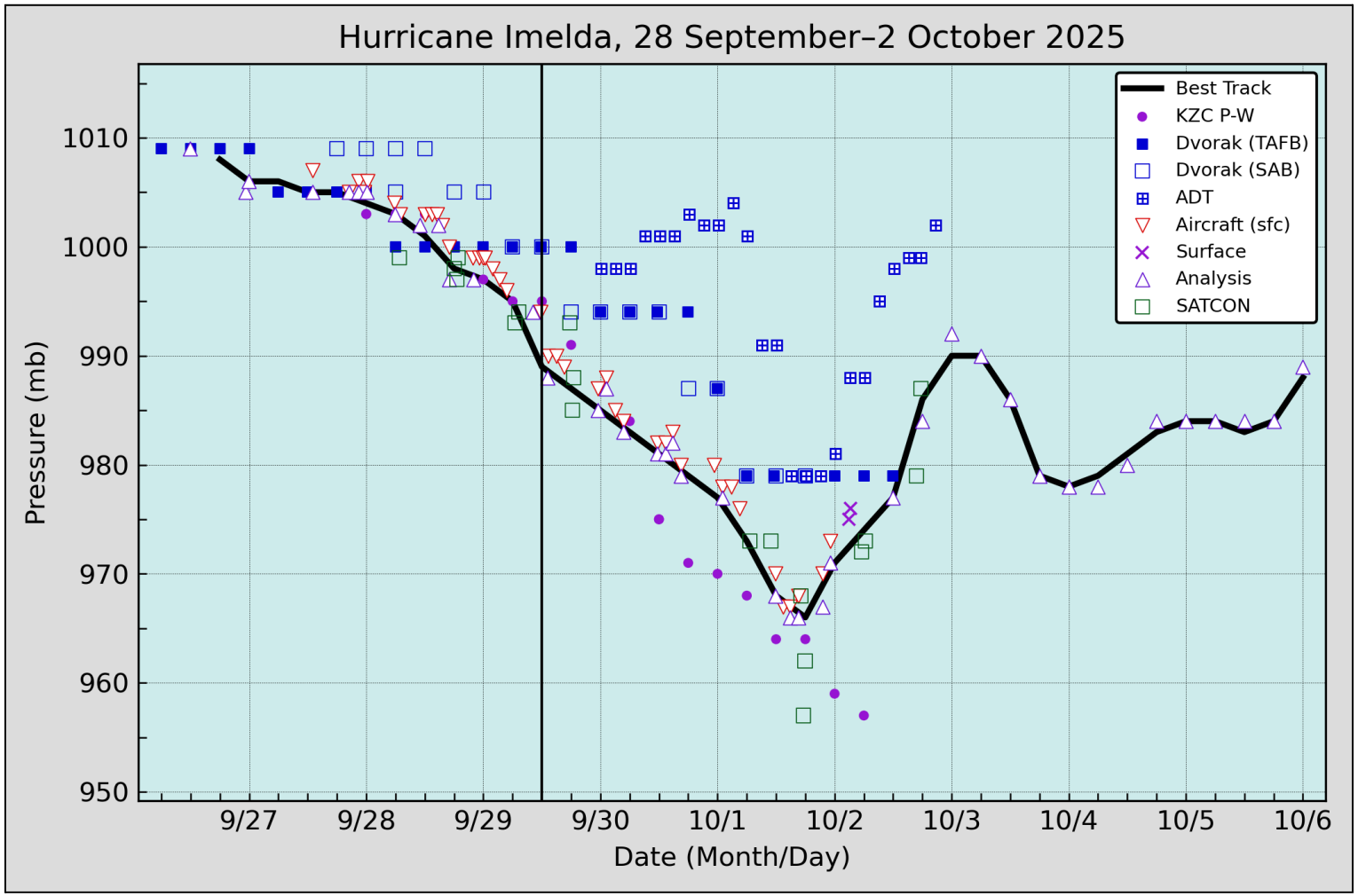


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Imelda, 28 September–2 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC, and the solid vertical line corresponds to landfall.

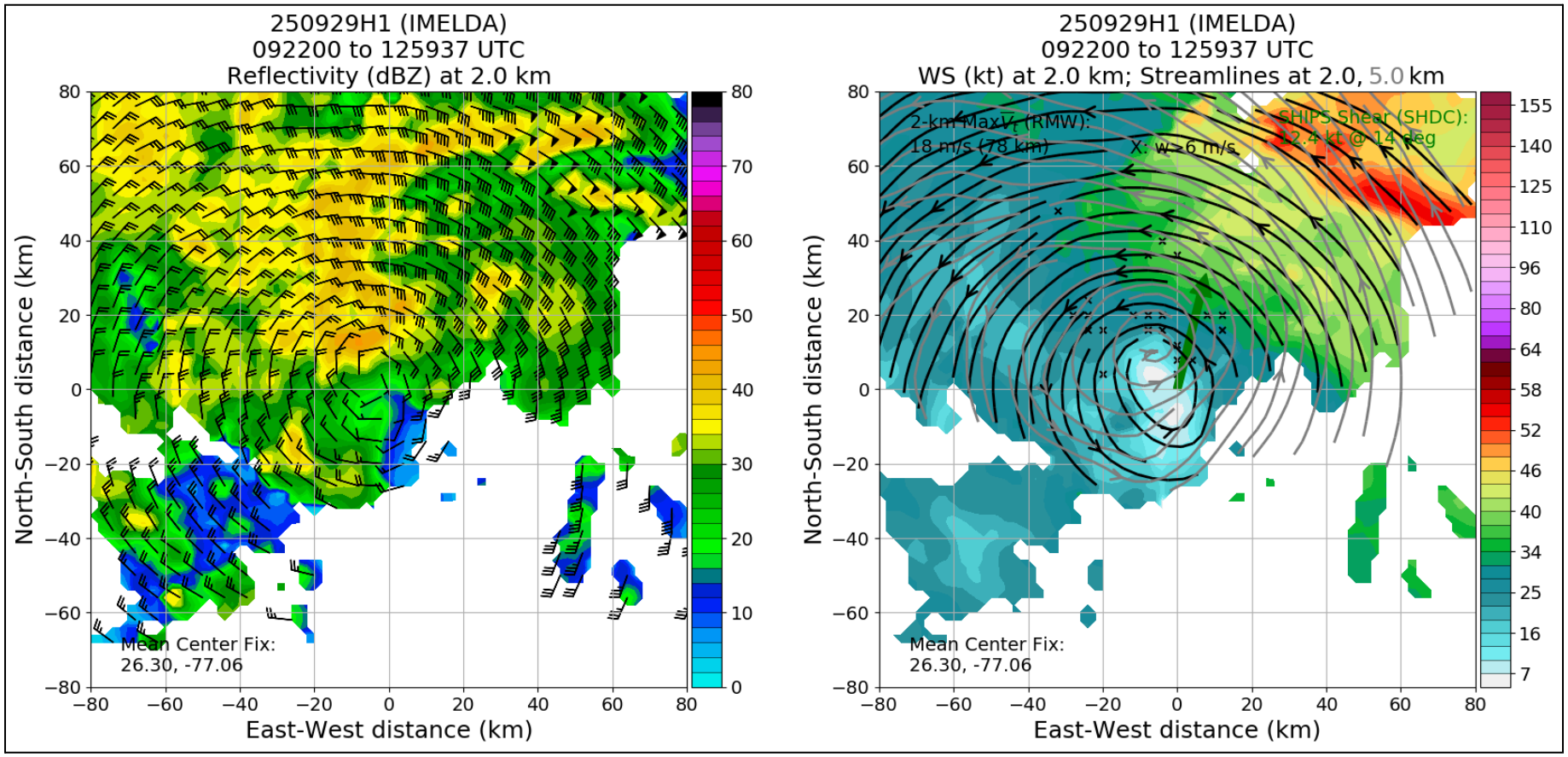


Figure 4. NOAA Tail Doppler Radar data of Imelda from 0922–1259 UTC 29 September. (Left) 2.0 km reflectivity image (dBZ) and winds (black barbs). (Right) 2.0 km wind speed (kt) along with streamlines at 2.0 km (black lines) and 5.0 km (gray lines), showing the vortex tilt with height due to southerly shear (green arrow).

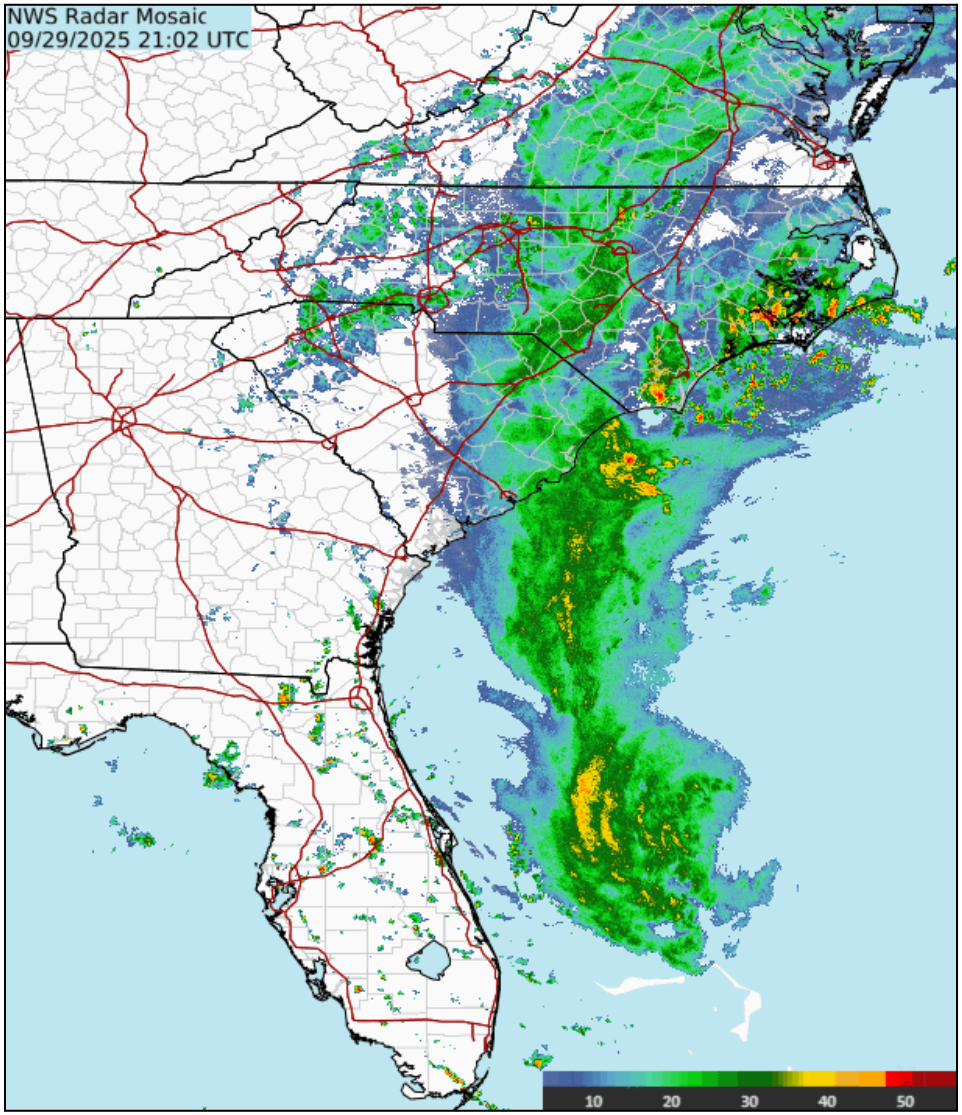


Figure 5. NWS radar mosaic image of Tropical Storm Imelda at 2102 UTC 29 September 2025 when it was located offshore of the east coast of Florida. Image courtesy of Brian McNoldy, University of Miami, Rosenstiel School.

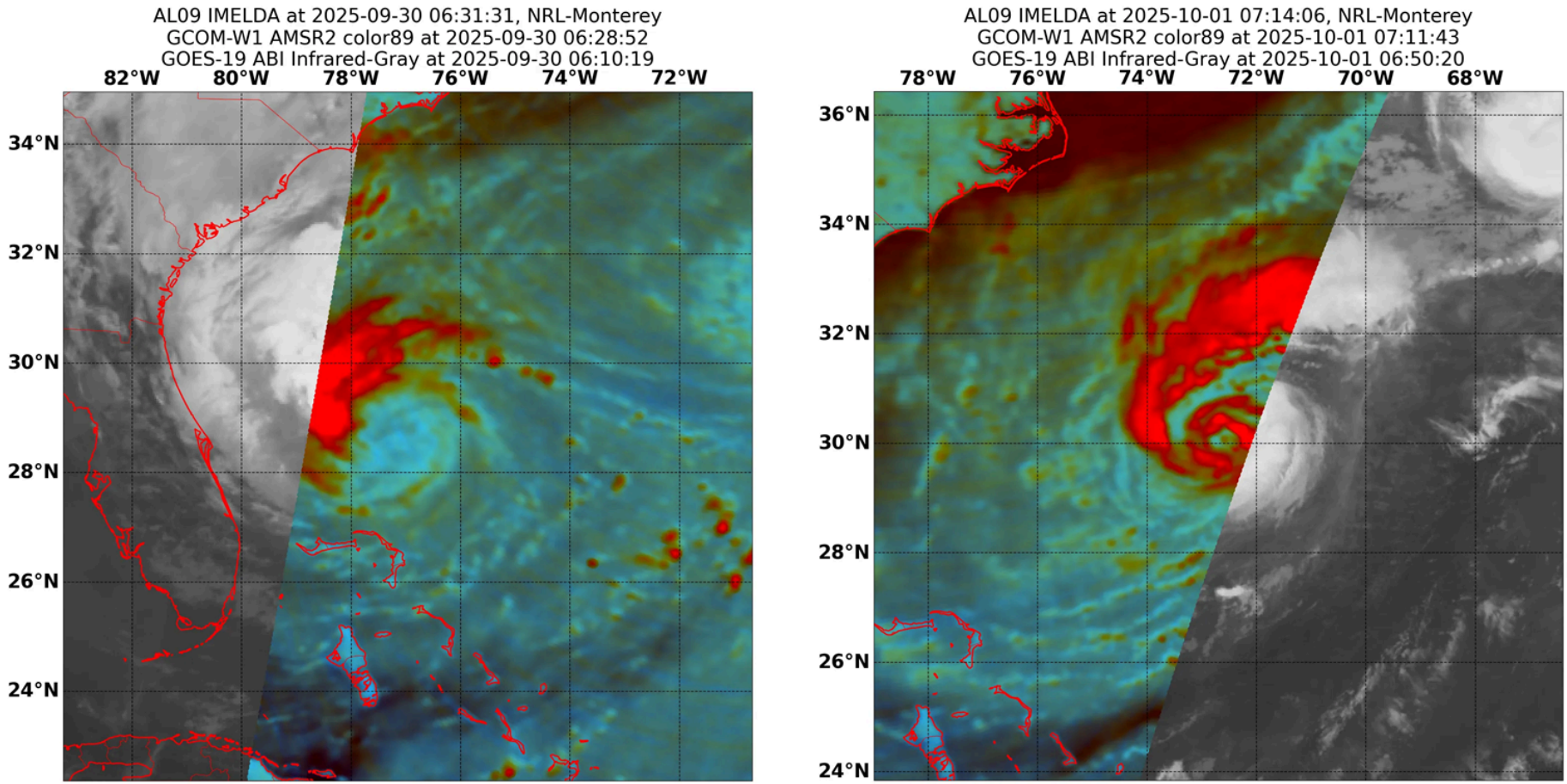


Figure 6. Passive microwave 89-GHz color composite images of Imelda at (left) 0628 UTC 30 September 2025 with an exposed low-level center, and (right) 0711 UTC 1 October 2025 with a ragged mid-level eyewall.

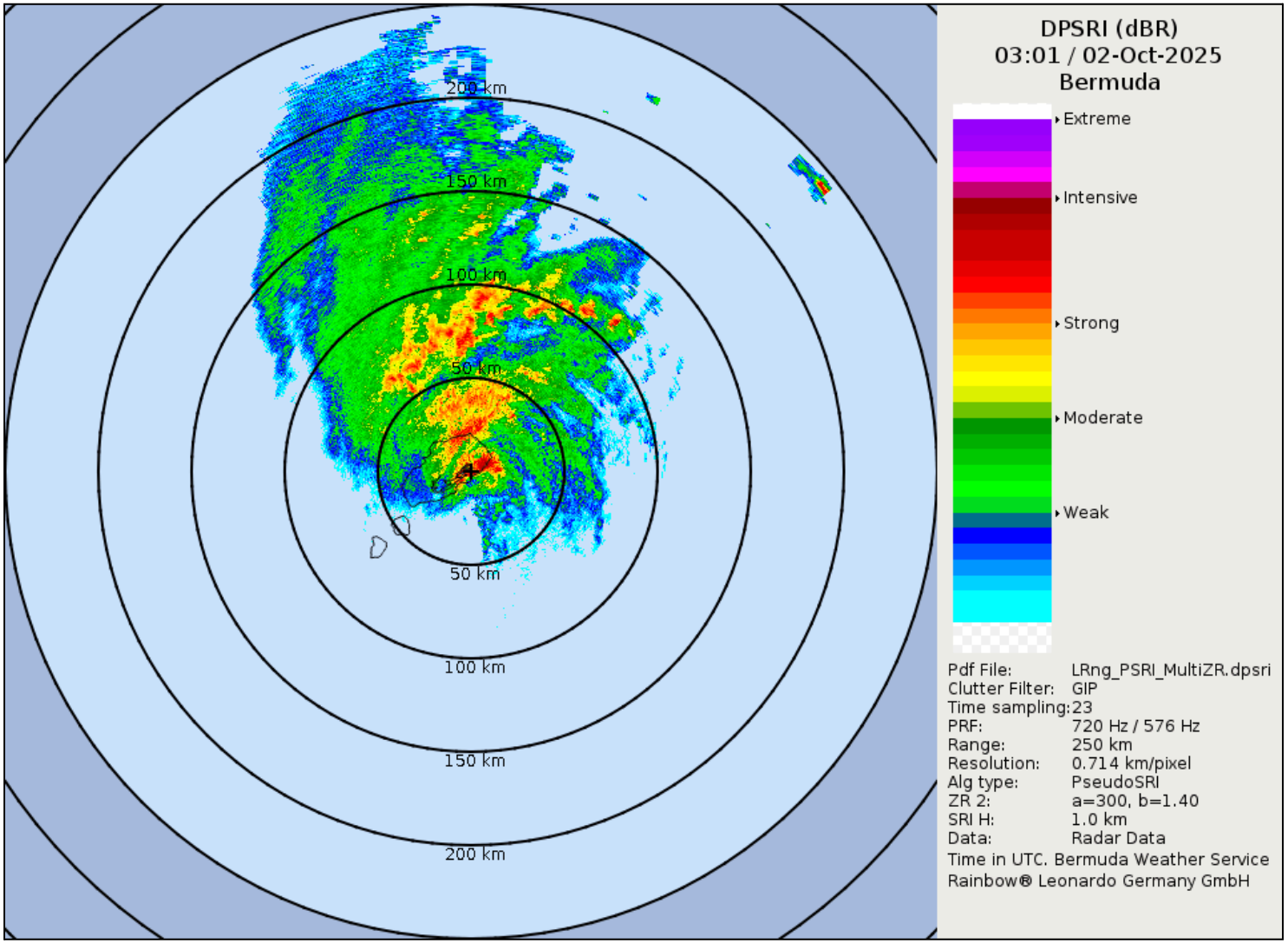


Figure 7. Radar image of Hurricane Imelda passing just south of Bermuda at 0301 UTC 2 October 2025. Data courtesy of the Bermuda Weather Service. Image courtesy of Brian McNoldy, University of Miami, Rosenstiel School.

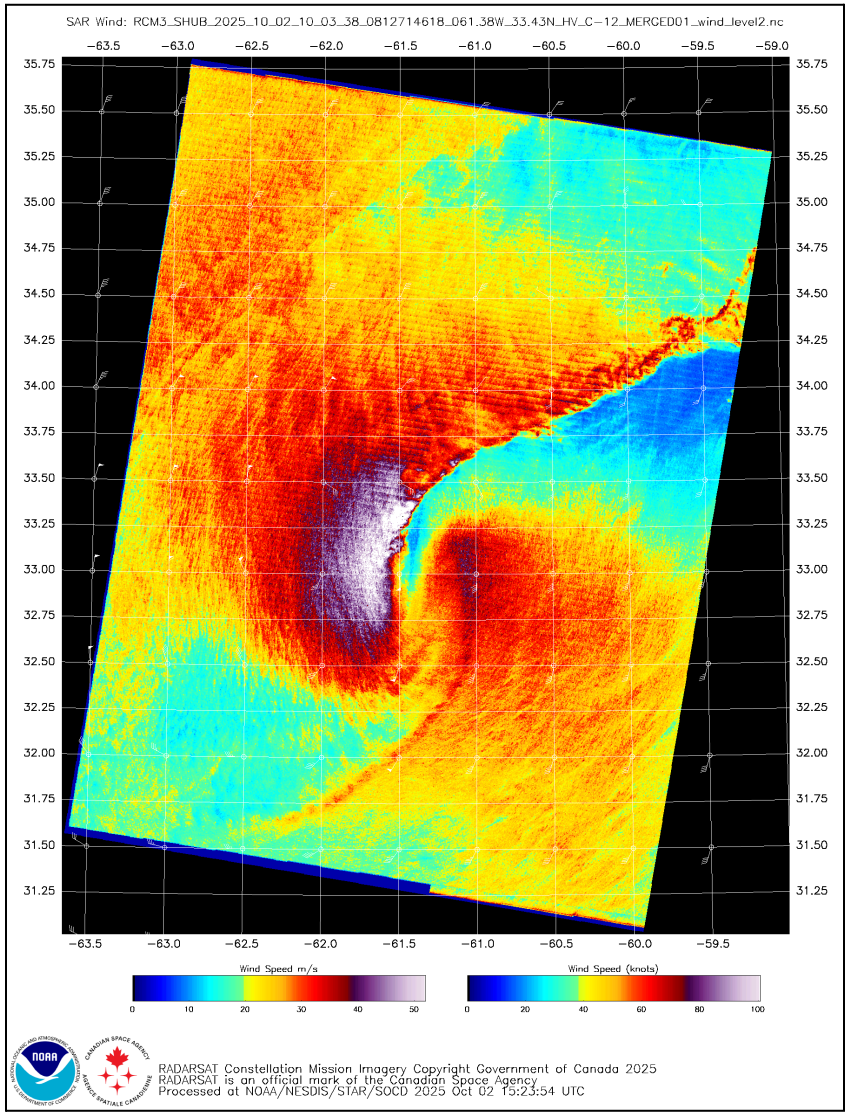


Figure 8. RCM-3 synthetic aperture radar (SAR) image of Imelda at 1003 UTC 2 October 2025, around the time that its extratropical transition was completed. Image courtesy NOAA/NESDIS/STAR.

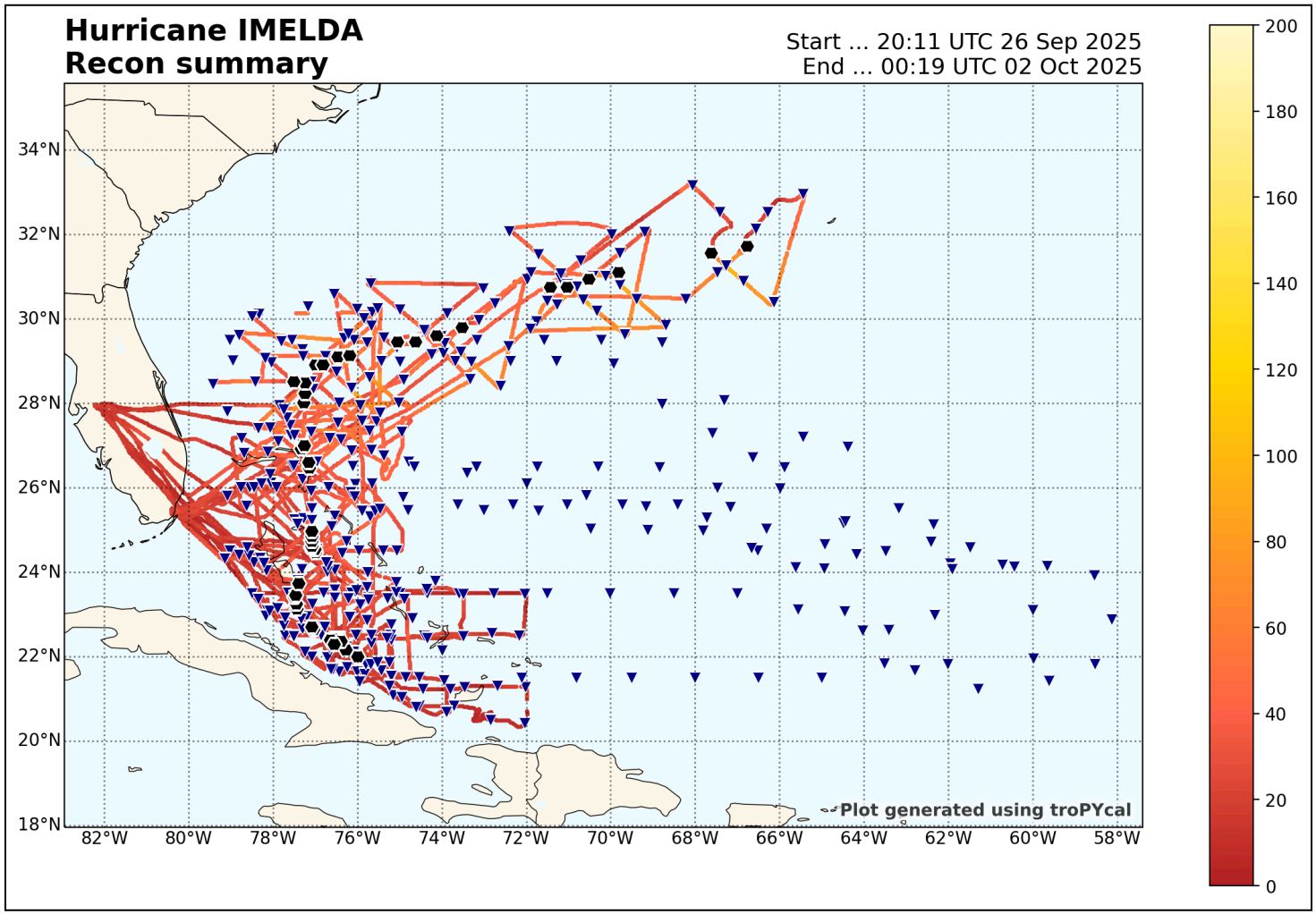
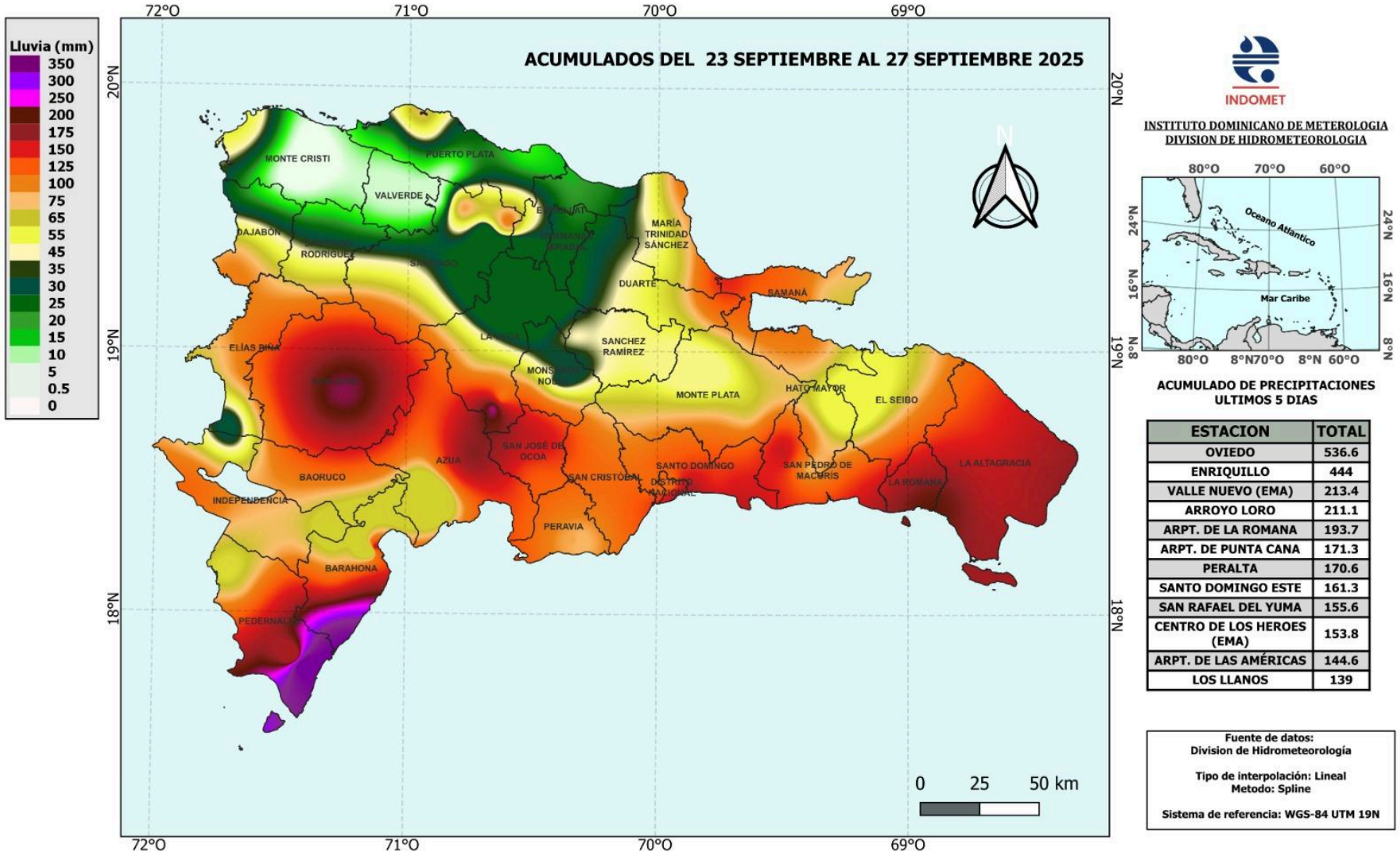


Figure 9. Air Force Reserve and NOAA Hurricane Hunter aircraft flight tracks (colored lines) from reconnaissance missions into Imelda from 26 September–2 October 2025. The black markers show center fix locations, and the blue triangles indicate dropsonde locations (including dropsondes from synoptic surveillance missions around both Imelda and Humberto). The color of the flight track represents the observed flight-level wind speed in knots at that location (see legend).



ESTACION	TOTAL
OVIEDO	536.6
ENRIQUILLO	444
VALLE NUEVO (EMA)	213.4
ARROYO LORO	211.1
ARPT. DE LA ROMANA	193.7
ARPT. DE PUNTA CANA	171.3
PERALTA	170.6
SANTO DOMINGO ESTE	161.3
SAN RAFAEL DEL YUMA	155.6
CENTRO DE LOS HEROES (EMA)	153.8
ARPT. DE LAS AMÉRICAS	144.6
LOS LLANOS	139

Figure 10. Dominican Republic rainfall totals (mm) from 23–27 September 2025 associated with the passage of the tropical wave from which Imelda developed. Image courtesy of the Dominican Institute of Meteorology (INDOMET).



Figure 11. Roof and structural damage to a former hotel on Langton Hill in the Pembroke Parish of Bermuda. Image courtesy Akil Simmons, *The Royal Gazette*.

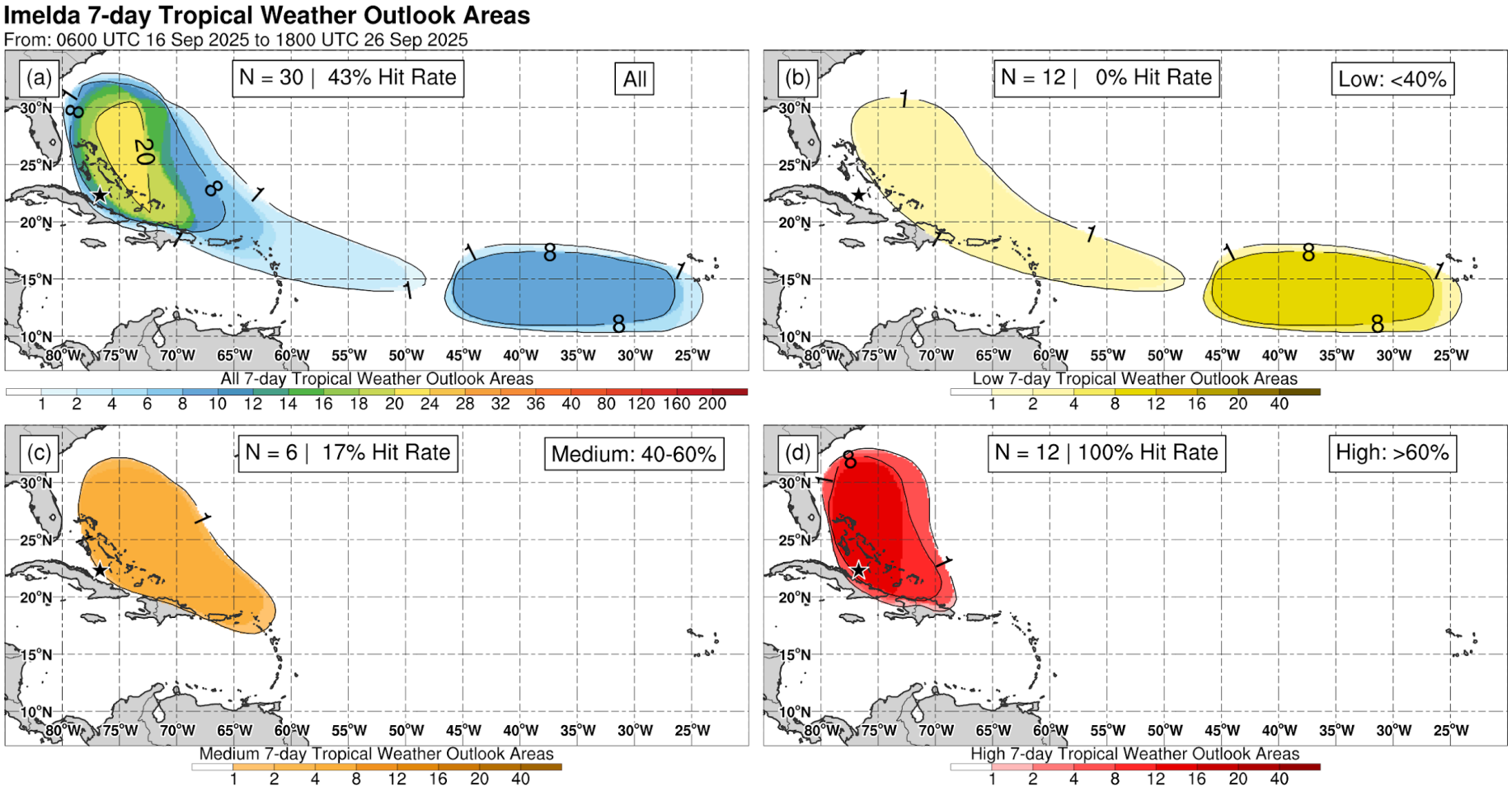


Figure 12. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Imelda for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.

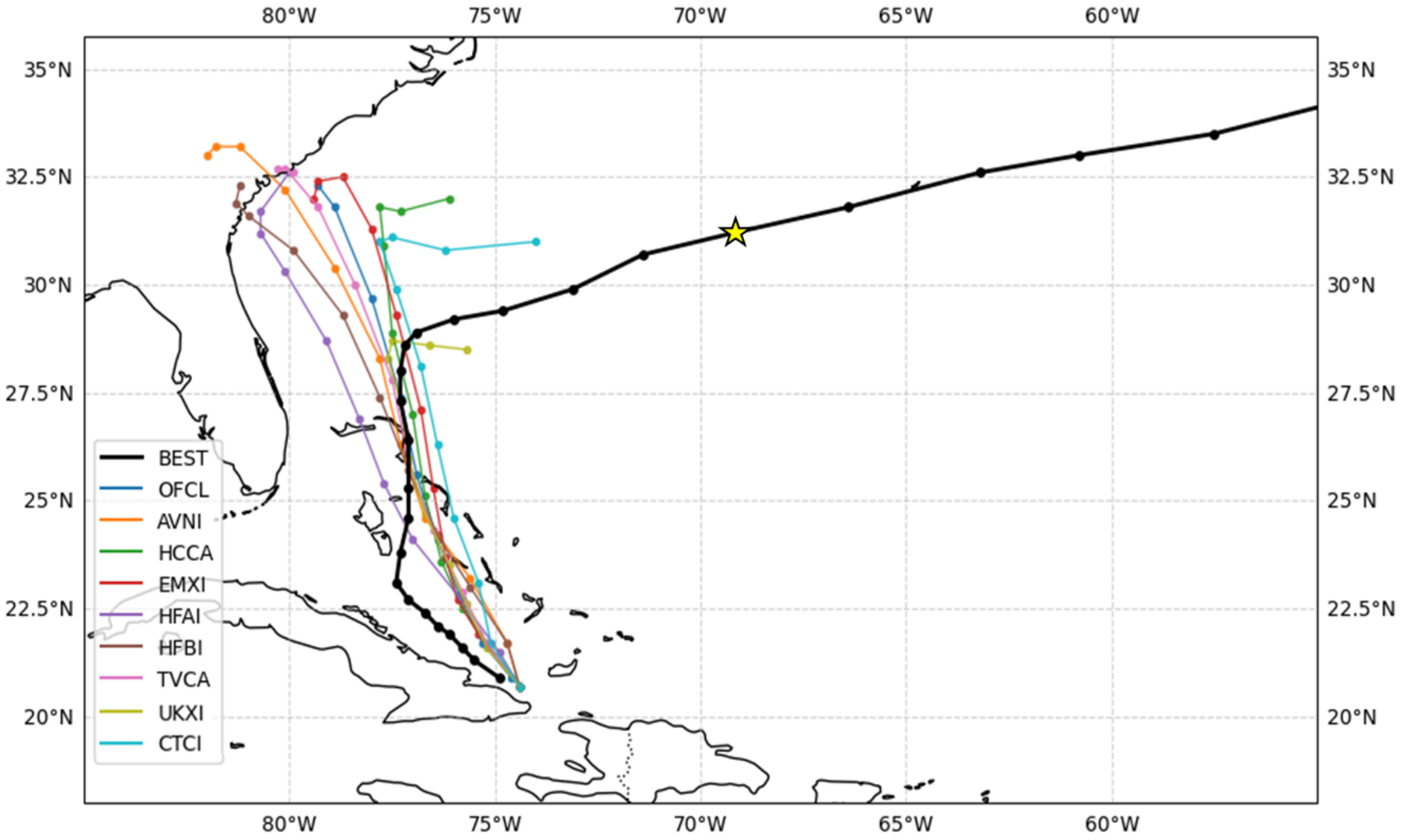


Figure 13. A comparison of the NHC official track forecast (OFCL) with various 5-day model forecasts at 1800 UTC 26 September 2025, when advisories were initiated on Imelda as a potential tropical cyclone. The best track is depicted by the black line, and the yellow star shows the verifying 5-day position of Imelda at 1800 UTC 1 October 2025.

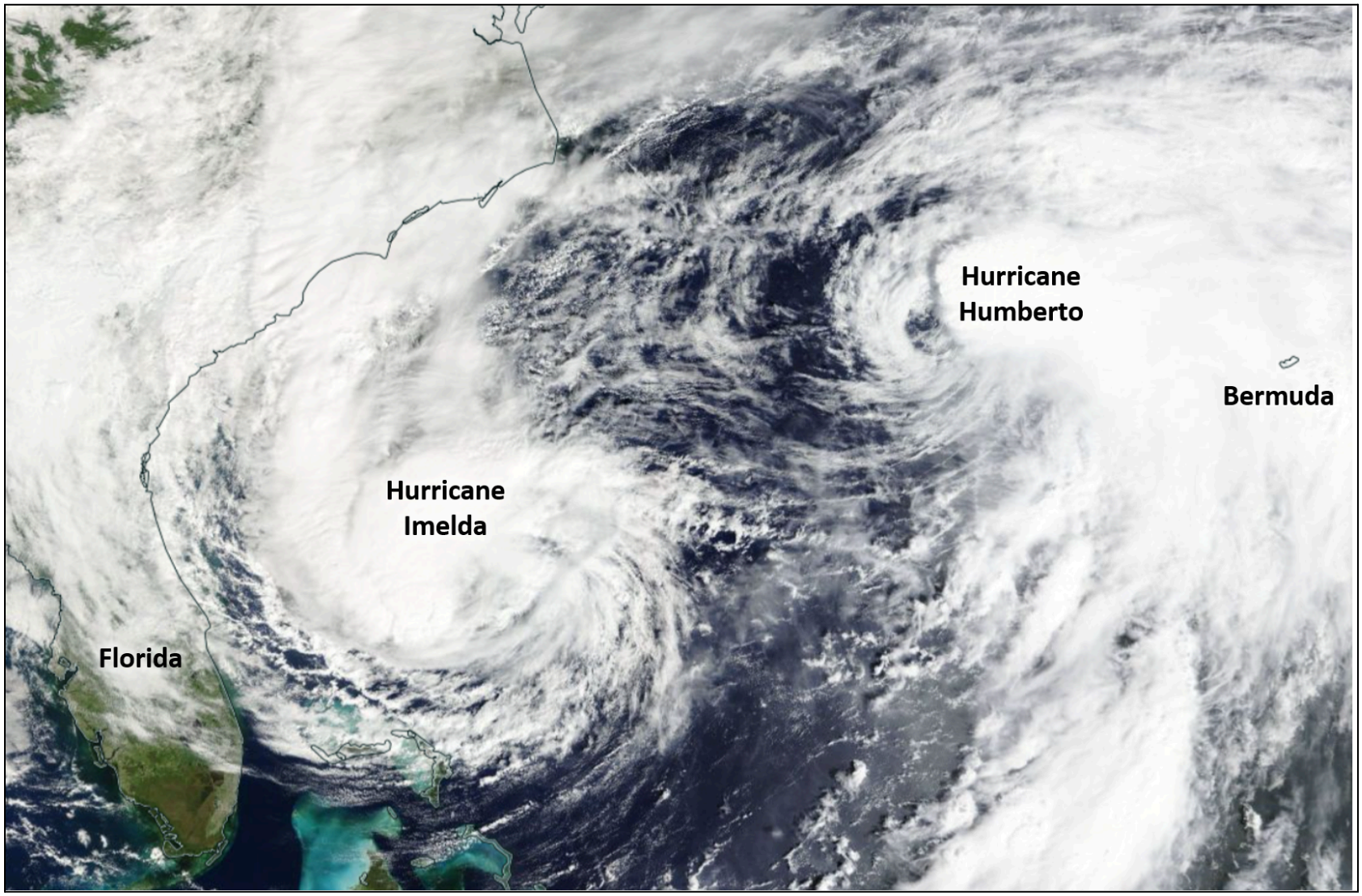


Figure 14. Terra MODIS true color image of the western Atlantic Ocean at 1425 UTC 30 September 2025, showing Hurricane Imelda and Hurricane Humberto in close proximity between Florida and Bermuda. Image courtesy NASA Worldview.

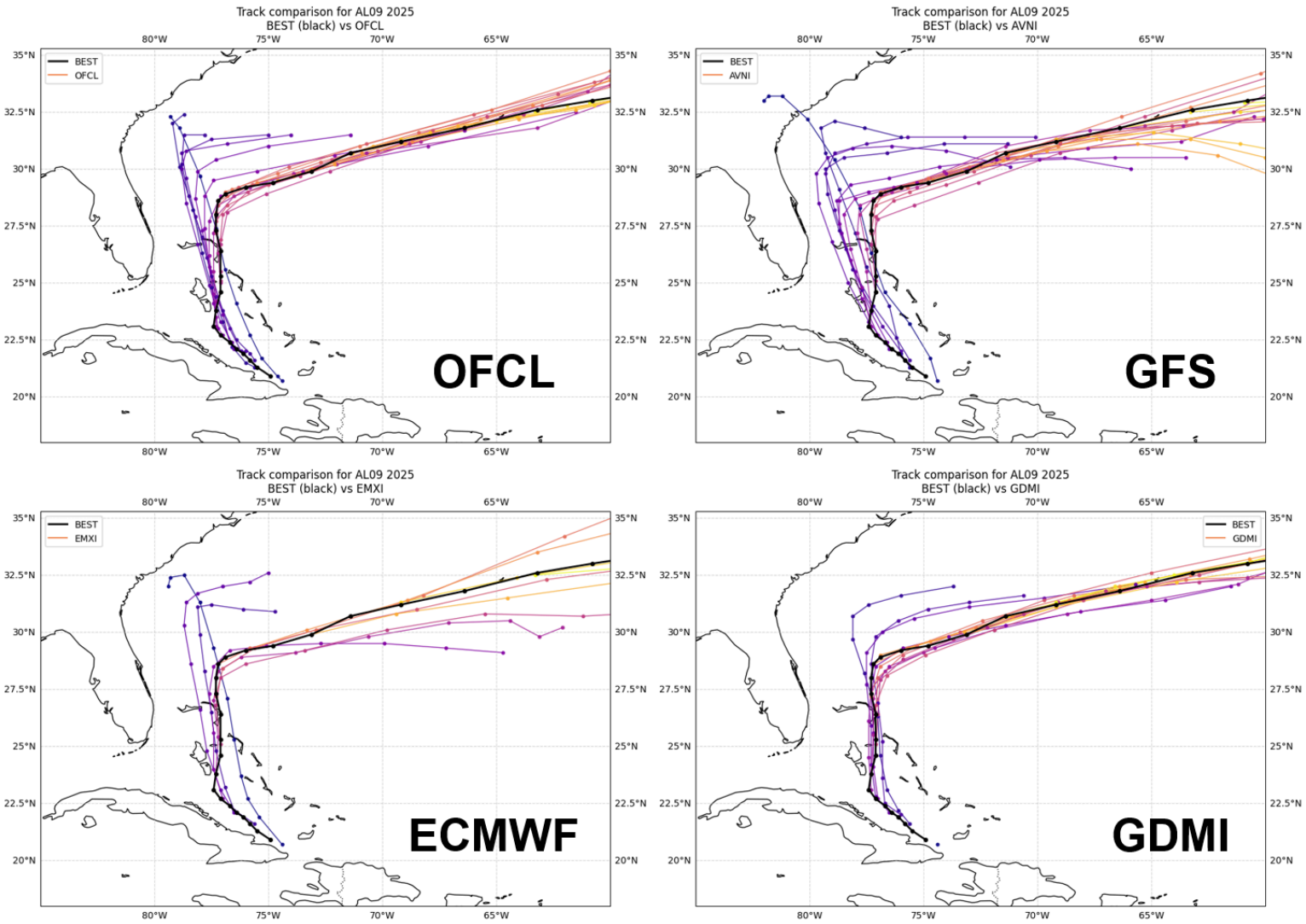


Figure 15. Model trend comparisons of Imelda track predictions from the NHC official forecast (top left), GFS (top right), ECMWF (bottom left), and Google DeepMind ensemble (bottom right). Earlier (later) forecasts are represented by cooler (warmer) colored lines. The best track is depicted by the black line.

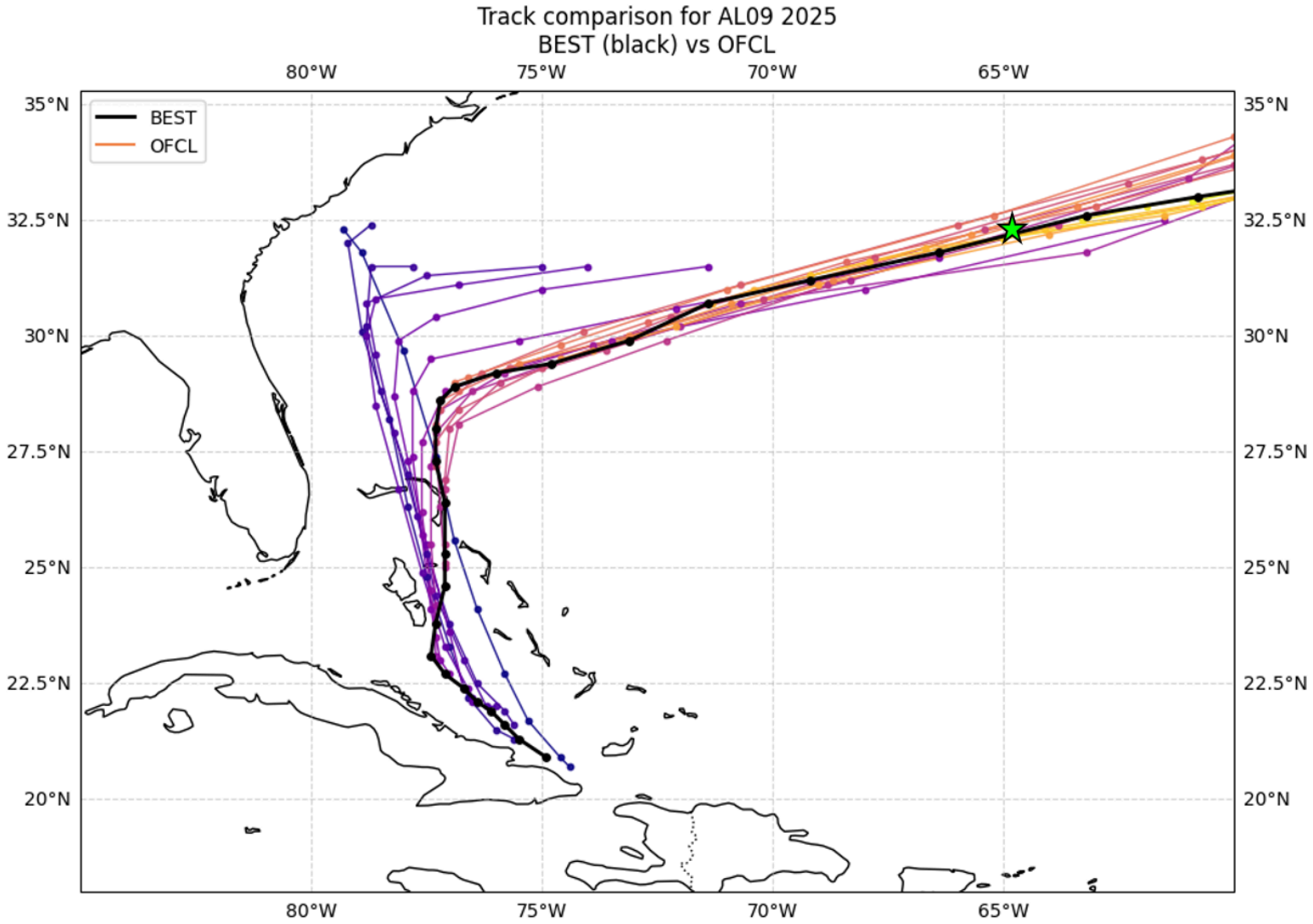


Figure 16. A composite of NHC official track forecasts for Imelda, with Bermuda highlighted by a green star. Earlier (later) forecasts are represented by cooler (warmer) colored lines. The best track is depicted by the black line.

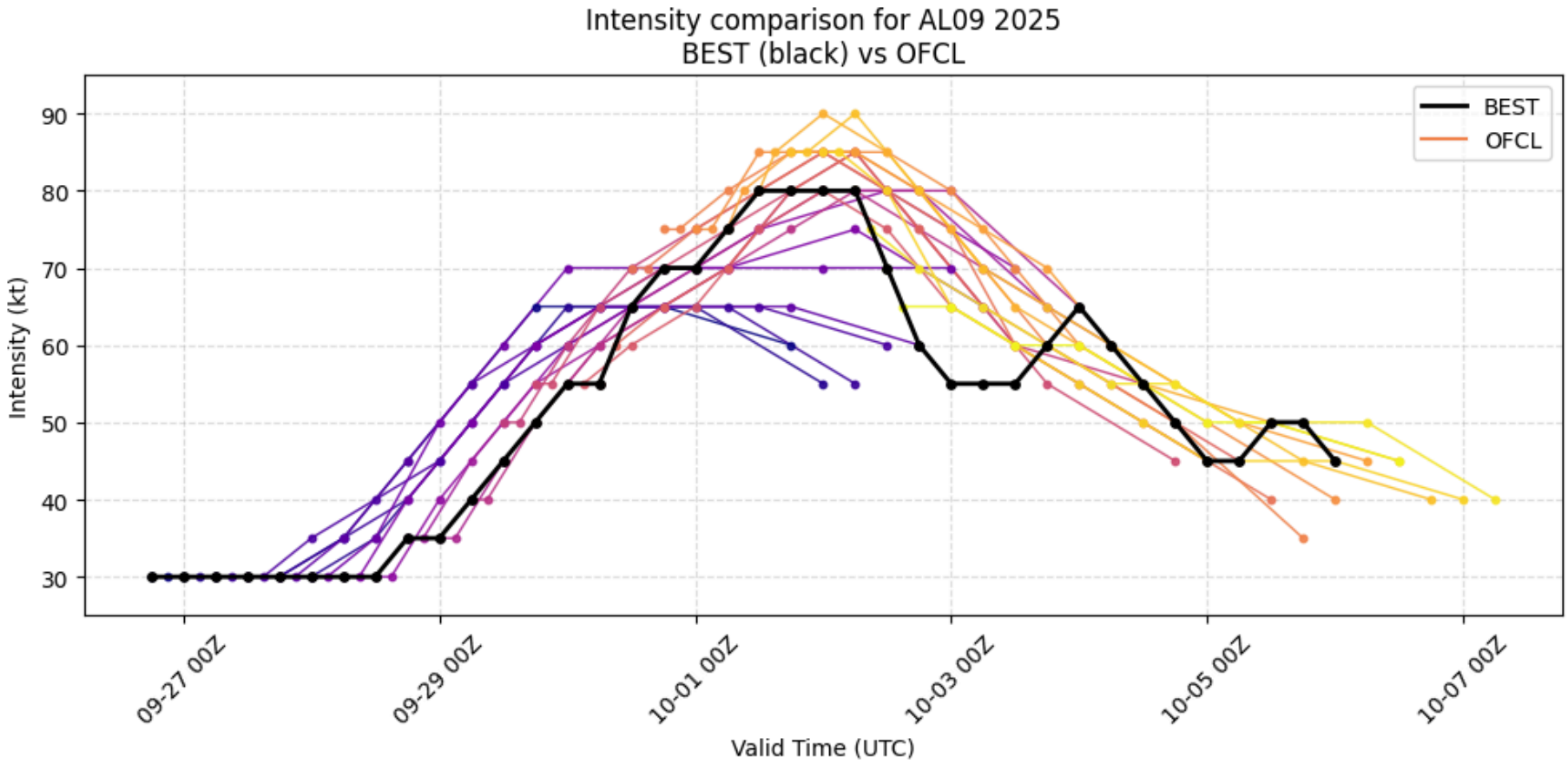


Figure 17. A composite of NHC official intensity forecasts for Imelda from 1800 UTC 26 September to 1200 UTC 2 October 2025. Earlier (later) forecasts are represented by cooler (warmer) colors. The best track is depicted by the black line.